



UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Advanced Level

MATHEMATICS

9709/31

Paper 3 Pure Mathematics 3 (P3)

May/June 2012

1 hour 45 minutes

Additional Materials: Answer Booklet/Paper
Graph Paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 75.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of **3** printed pages and **1** blank page.



- 1 Solve the equation $|4 - 2^x| = 10$, giving your answer correct to 3 significant figures. [3]

- 2 (i) Expand $\frac{1}{\sqrt{1-4x}}$ in ascending powers of x , up to and including the term in x^2 , simplifying the coefficients. [3]

- (ii) Hence find the coefficient of x^2 in the expansion of $\frac{1+2x}{\sqrt{4-16x}}$. [2]

- 3 The polynomial $p(x)$ is defined by

$$p(x) = x^3 - 3ax + 4a,$$

where a is a constant.

- (i) Given that $(x - 2)$ is a factor of $p(x)$, find the value of a . [2]

- (ii) When a has this value,

- (a) factorise $p(x)$ completely, [3]

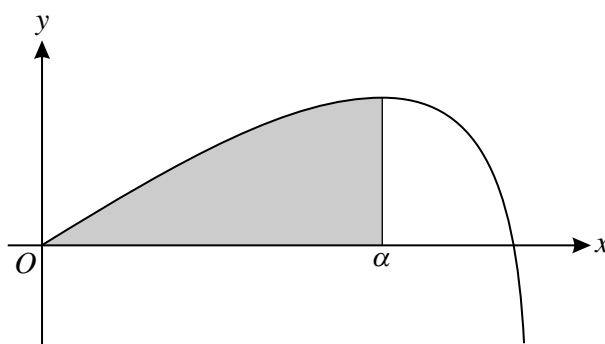
- (b) find all the roots of the equation $p(x^2) = 0$. [2]

- 4 The complex number u is defined by $u = \frac{(1+2i)^2}{2+i}$.

- (i) Without using a calculator and showing your working, express u in the form $x + iy$, where x and y are real. [4]

- (ii) Sketch an Argand diagram showing the locus of the complex number z such that $|z - u| = |u|$. [3]

5



The diagram shows the curve

$$y = 8 \sin \frac{1}{2}x - \tan \frac{1}{2}x$$

for $0 \leq x < \pi$. The x -coordinate of the maximum point is α and the shaded region is enclosed by the curve and the lines $x = \alpha$ and $y = 0$.

- (i) Show that $\alpha = \frac{2}{3}\pi$. [3]

- (ii) Find the exact value of the area of the shaded region. [4]

6 The equation of a curve is $3x^2 - 4xy + y^2 = 45$.

(i) Find the gradient of the curve at the point $(2, -3)$. [4]

(ii) Show that there are no points on the curve at which the gradient is 1. [3]

7 The variables x and y are related by the differential equation

$$\frac{dy}{dx} = \frac{6xe^{3x}}{y^2}.$$

It is given that $y = 2$ when $x = 0$. Solve the differential equation and hence find the value of y when $x = 0.5$, giving your answer correct to 2 decimal places. [8]

8 The point P has coordinates $(-1, 4, 11)$ and the line l has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$.

(i) Find the perpendicular distance from P to l . [4]

(ii) Find the equation of the plane which contains P and l , giving your answer in the form $ax + by + cz = d$, where a, b, c and d are integers. [5]

9 By first expressing $\frac{4x^2 + 5x + 3}{2x^2 + 5x + 2}$ in partial fractions, show that

$$\int_0^4 \frac{4x^2 + 5x + 3}{2x^2 + 5x + 2} dx = 8 - \ln 9. \quad [10]$$

10 (i) It is given that $2 \tan 2x + 5 \tan^2 x = 0$. Denoting $\tan x$ by t , form an equation in t and hence show that either $t = 0$ or $t = \sqrt[3]{t + 0.8}$. [4]

(ii) It is given that there is exactly one real value of t satisfying the equation $t = \sqrt[3]{t + 0.8}$. Verify by calculation that this value lies between 1.2 and 1.3. [2]

(iii) Use the iterative formula $t_{n+1} = \sqrt[3]{t_n + 0.8}$ to find the value of t correct to 3 decimal places. Give the result of each iteration to 5 decimal places. [3]

(iv) Using the values of t found in previous parts of the question, solve the equation

$$2 \tan 2x + 5 \tan^2 x = 0$$

for $-\pi \leq x \leq \pi$. [3]

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.